

Soluciones - Certamen 2

June 5, 2025

Problem 1

- ① (**7 points**) A pointlike nonrelativistic particle of mass m scatters in the time-dependent spherically symmetric harmonic potential

$$U(r, t) = \frac{k(t) r^2}{2}, \quad k(t) = \frac{\kappa}{t^2}, \quad \kappa > 0. \quad (1)$$

- ① (**2 points**) Identify all continuous symmetries of the action. Analyze if there are symmetries of type

$$\mathbf{r} \rightarrow \alpha \mathbf{r}, \quad t \rightarrow \beta t, \quad \alpha, \beta = \text{const}$$

which do not change the equations of motion.

- ② (**3 points**) Construct the integrals of motion which might exist according to Noether's theorem. Demonstrate that these quantities are conserved using the equations of motion. Note that the lagrangian of the 3D harmonic oscillator in Cartesian coordinates can be represented as a sum of 3 independent pieces,

$$L = \sum_{a=1}^3 \left(\frac{m \dot{x}_a^2}{2} - \frac{k(t) x_a^2}{2} \right),$$

Problem 1 (part (c))

- ① (7 points) A pointlike nonrelativistic particle of mass m scatters in the time-dependent spherically symmetric harmonic potential

$$U(r, t) = \frac{k(t) r^2}{2}, \quad k(t) = \frac{\kappa}{t^2}, \quad \kappa > 0. \quad (1)$$

- (c) (2 points) We have seen in lectures that in static (time-independent) spherical harmonic oscillator the particle's trajectory is a plain ellipse. Analyze if this property remains valid in the time-dependent potential given in Eq. (1).

Problem 1 (derivation of symmetry)

We have symmetry w.r.t. rotation around any axis (spherical symmetry),
⇒ all components of angular momentum vector $\mathbf{M}$ are conserved

Explicit time, coordinate dependence of L
⇒ Energy, linear momentum $\mathbf{P}$ are NOT conserved

Problem 1 (derivation of symmetry)

Have also “rescaling” symmetry

$$\mathbf{r} \rightarrow \alpha \mathbf{r}, \quad t \rightarrow \beta t, \quad \alpha, \beta \in \mathbb{R}$$

The action transforms as

$$S \rightarrow S' = \int dt \left(\frac{\alpha^2}{\beta} \frac{m \dot{\mathbf{r}}^2}{2} - \frac{\alpha^2}{\beta} \frac{\kappa r^2}{2t^2} \right) = \frac{\alpha^2}{\beta} \int dt \left(\frac{m \dot{\mathbf{r}}^2}{2} - \frac{\kappa r^2}{2t^2} \right) = \frac{\alpha^2}{\beta} S$$

- We should NOT forget extra “ β ” which comes from $\int dt$
- The equations of motion remain invariant for any choice of α, β , however the action is invariant only if

$$\frac{\alpha^2}{\beta} = 1, \quad \alpha = \sqrt{\beta}$$

- Only in the latter case we can apply the Noether theorem and construct the integrals of motion associated with this symmetry.

Problem 1 (proof of invariance)

We analyzed symmetry

$$\mathbf{r} \rightarrow \alpha \mathbf{r}, \quad t \rightarrow \alpha^2 t, \quad \alpha \in \mathbb{R}$$

in Lecture 7, so we repeat almost literally those results. However, now energy $E \neq \text{const}$

$$S = \int dt \left(\frac{m \dot{\mathbf{r}}^2}{2} - \frac{\kappa r^2}{2t^2} \right)$$

Change of different quantities under rescaling:

$$\mathbf{r} \rightarrow \alpha \mathbf{r}, \quad t \rightarrow \alpha^2 t, \quad \alpha = e^\lambda \approx 1 + \lambda \quad (1)$$

$$\frac{m \dot{\mathbf{r}}^2}{2} \rightarrow \frac{1}{\alpha^2} \frac{m \dot{\mathbf{r}}^2}{2}, \quad U(r) \rightarrow \frac{1}{\alpha^2} U(r)$$

$$\Rightarrow S = \int dt L \rightarrow \int \cancel{\alpha^2} dt \frac{1}{\cancel{\alpha^2}} L = \text{const}$$

This proves that (1) is indeed a continuous symmetry of the action.

Problem 1 (derivation of invariant)

To find the associated Noether invariant, consider $\alpha = \alpha(t) = e^{\lambda(t)}$, $|\lambda| \ll 1$ and pick up $\mathcal{O}(\dot{\lambda})$ term in the expansion of the action:

$$d\mathbf{r} \rightarrow (d\mathbf{r} + \mathbf{r}d\lambda) = (d\mathbf{r} + \mathbf{r}\dot{\lambda}dt), \quad dt \rightarrow (1 + 2\dot{\lambda}t) dt$$

$$\mathbf{v} \equiv \frac{d\mathbf{r}}{dt} \rightarrow \frac{\mathbf{v} + \mathbf{r}\dot{\lambda}}{1 + 2\dot{\lambda}t} \approx \mathbf{v} + \dot{\lambda}(\mathbf{r} - 2t\mathbf{v})$$

$$\begin{aligned} S &\equiv \int dt L \rightarrow \int dt L (1 + 2\dot{\lambda}t) + \int dt m \mathbf{v} \cdot (\mathbf{r} - 2t\mathbf{v}) \dot{\lambda} \\ &= \int dt [m \mathbf{v} \cdot (\mathbf{r} - 2t\mathbf{v}) + 2tL] \dot{\lambda} \end{aligned}$$

Problem 1 - Exam 2

$$\begin{aligned}
 Q &= \frac{\partial \delta L}{\partial \dot{\lambda}} = m \mathbf{v} \cdot (\mathbf{r} - 2t \mathbf{v}) + 2tL = \\
 &= m \mathbf{v} \cdot (\mathbf{r} - 2t \mathbf{v}) + 2t \left(\frac{m \mathbf{v}^2}{2} - U(r, t) \right) \\
 &= \mathbf{p} \cdot \mathbf{r} - 2 \left(\frac{m \mathbf{v}^2}{2} + U(r, t) \right) t
 \end{aligned}$$

– Check that $Q = \text{const}$ using equations of motion and explicit expression for potential:

$$\begin{aligned}
 \frac{dQ}{dt} &= \dot{\mathbf{p}} \cdot \mathbf{r} + \mathbf{p} \cdot \mathbf{v} - 2 \left(\frac{m \mathbf{v}^2}{2} + U(r, t) \right) - 2t \left(\frac{\mathbf{p} \cdot \dot{\mathbf{p}}}{m} + \frac{\partial U(r, t)}{\partial t} + \mathbf{v} \cdot \frac{\partial U(r, t)}{\partial \mathbf{r}} \right) \\
 &= -\frac{\partial U}{\partial \mathbf{r}} \cdot \mathbf{r} + \cancel{\mathbf{p} \cdot \mathbf{v}} - 2 \left(\cancel{\frac{m \mathbf{v}^2}{2}} + U(r, t) \right) - \\
 &\quad - 2t \left(-\frac{1}{m} \cancel{\mathbf{p} \cdot \frac{\partial U(r, t)}{\partial \mathbf{r}}} + \frac{\partial U(r, t)}{\partial t} + \cancel{\mathbf{v} \cdot \frac{\partial U}{\partial \mathbf{r}}} \right) = \left[-\mathbf{r} \cdot \frac{\partial}{\partial \mathbf{r}} - 2 - 2t \frac{\partial}{\partial t} \right] U \\
 &= [-2 - 2 - 2 \cdot (-2)] U(r, t) = 0
 \end{aligned}$$

as expected (!!!)

Problem 1 - Exam 2

The lagrangian of the 3D harmonic oscillator in Cartesian coordinates can be represented as a sum of 3 independent pieces,

$$L = \sum_{a=1}^3 \left(\frac{m \dot{x}_a^2}{2} - \frac{k(t) x_a^2}{2} \right),$$

for this reason:

- 1 The equations of motion in Cartesian coordinates decouple from each other,

$$m\ddot{x} = -\frac{\kappa}{t^2}x, \quad m\ddot{y} = -\frac{\kappa}{t^2}y, \quad m\ddot{z} = -\frac{\kappa}{t^2}z, \quad (1)$$

We can analyze separately each of the 3 pieces and obtain that there is conservation of 3 invariants

$$Q_a = p_a x_a - 2 \left(\frac{m v_a^2}{2} + \frac{k(t) x_a^2}{2} \right) t = m \dot{x}_a (x_a - t \dot{x}_a) - \frac{\kappa x_a^2}{t}, \quad a = x, y, z$$

however they are NOT completely independent from what we found earlier:

- 1 If we start from Q_x and apply rotation by angle $\pi/2$ around axis y or z , we'll get Q_z, Q_y

Problem 1 - Exam 2

$$m\ddot{x} = -\frac{\kappa}{t^2}x, \quad m\ddot{y} = -\frac{\kappa}{t^2}y, \quad m\ddot{z} = -\frac{\kappa}{t^2}z, \quad (3)$$

$$Q_a = p_a \dot{x}_a - 2 \left(\frac{m v_a^2}{2} + \frac{k(t) x_a^2}{2} \right) t = m \dot{x}_a (x_a - t \dot{x}_a) - \frac{\kappa x_a^2}{t}, \quad a = x, y, z$$

- ① The equation (3) is the Euler's equation, has solutions $x_a = \mathcal{A}_a t^{\alpha_1} + \mathcal{B}_a t^{\alpha_2}$ where $\alpha_{1,2}$ are roots of

$$\alpha(\alpha - 1) + \frac{\kappa}{m} = 0, \quad \alpha_{1,2} = \frac{1 \pm \sqrt{1 - 4\kappa/m}}{2}, \quad \alpha_1 + \alpha_2 = 1$$

and $\mathcal{A}_a, \mathcal{B}_a$ are arbitrary constants. Can check that

$Q_a = \mathcal{A}_a \mathcal{B}_a (m - 4\kappa) = \text{const}$, i.e. the t -dependence cancels

If we consider $\mathcal{A}_a, \mathcal{B}_a$ as Cartesian components of 3D vector, then can see that 3D solution is given by a sum of 2 contributions

$$\mathbf{x} = \mathcal{A} t^{\alpha_1} + \mathcal{B} t^{\alpha_2}$$

and thus always remains in a plane formed by vectors $\mathcal{A}$ and $\mathcal{B}$

Problem 1 - Exam 2

- ① Since $\mathbf{M} = \text{const}$, repeating discussion from Lecture 7, conclude that trajectory always remains in plane $\perp$ to vector $\mathbf{M}$.

① Using explicit solution from the previous problem, can get

$$\mathbf{x} = \mathcal{A} t^{\alpha_1} + \mathcal{B} t^{\alpha_2}, \quad \dot{\mathbf{x}} = \alpha_1 \mathcal{A} t^{\alpha_1-1} + \alpha_2 \mathcal{B} t^{\alpha_2-1}$$

$$\mathbf{M} = m\mathbf{x} \times \dot{\mathbf{x}} = m\mathcal{A} \times \mathcal{B} (\alpha_1 - \alpha_2) = m\mathcal{A} \times \mathcal{B} \sqrt{1 - 4\kappa/m}$$

- ② Since $|\mathbf{M}| = \text{const}$, repeating discussion from Lecture 7, conclude that the 2nd law of Kepler remains valid
- ③ The first and the third laws of Kepler are NOT valid, since they use explicit trajectory for their proof.

Problem 2

- ❶ (8 **points**) In the lectures 12-13 we discussed that the covariant equations of motion for the massive particle in the electromagnetic field A^μ may be obtained from the action

$$S = \int d\tau \Lambda, \quad \Lambda = -m\sqrt{\dot{x}_\mu \dot{x}^\mu} + e\dot{x}_\mu A^\mu(x) \quad (1)$$

where τ is the proper time, $x^\mu(\tau)$ is the trajectory of the particle, and $\dot{x}^\mu = dx^\mu/d\tau \equiv u^\mu$ is the 4-velocity of the particle. Assume that we replaced the covariant lagrangian (1) with

$$\tilde{\Lambda} = \frac{m\dot{x}_\mu \dot{x}^\mu}{2} + e\dot{x}_\mu A^\mu(x) \quad (2)$$

- ❶ (2 **points**) Demonstrate that the action S is covariant under transformations of the Lorentz group when we use the lagrangian (2).
- ❷ (2 **points**) Write out the generalized momentum $\mathcal{P}_\mu = \partial\tilde{\Lambda}/\partial\dot{x}^\mu$ and the equations of motion which correspond to the lagrangian (2).

Problem 2 (part 2)

- (c) (**2 points**) Demonstrate that your equations of motion are equivalent to the equations of motion found in lectures 12,13 using the lagrangian (2). Demonstrate that the same equations can be obtained with any lagrangian $\bar{\Lambda}$ given by

$$\bar{\Lambda} = mf(\dot{x}_\mu \dot{x}^\mu) + e\dot{x}_\mu A^\mu(x)$$

where $f(y)$ is an arbitrary function which satisfies $\partial f(y)/\partial y = 1/2$ at $y = 1$ [*clearly, the lagrangians (1,2) correspond to $f(y) = \sqrt{y}$ and $f(y) = y/2$, respectively*].

- (d) (**2 points**) Find the nonrelativistic lagrangian L which corresponds to the lagrangian (2). Pay attention that the nonrelativistic lagrangian L contributes to the action as $S = \int dt L$.

Problem 2 (solution)

$$S = \int d\tau \tilde{\Lambda}, \quad \tilde{\Lambda} = \frac{m\dot{x}_\mu\dot{x}^\mu}{2} + e\dot{x}^\mu A_\mu(x), \quad \dot{x}^\mu = dx^\mu/d\tau \quad (1)$$

- ① The action S is invariant under transformations of the Lorentz group because we construct it from covariant objects $\dot{x}^\mu$, A^μ and $\dot{x}_\mu$, and after contraction we do not have “free” Lorentz indices. Can write out explicit transformations to show this: e.g.

$$\dot{x}^\mu \rightarrow \Lambda^\mu_\nu \dot{x}^\nu, \quad \dot{x}_\mu \rightarrow \Lambda_\mu^\alpha \dot{x}_\alpha, \quad \dot{x}^\mu \dot{x}_\mu \rightarrow \underbrace{\Lambda^\mu_\nu \Lambda_\mu^\alpha}_{\delta^\alpha_\nu} \dot{x}^\nu \dot{x}_\alpha = \dot{x}^\nu \dot{x}_\nu$$

(here Λ^μ_ν , Λ_μ^α are Lorentz boost matrices, don't confuse with invariant lagrangian Λ)

- ② The generalized momentum $\mathcal{P}_\mu = \partial\tilde{\Lambda}/\partial\dot{x}^\mu = m\dot{x}_\mu + eA_\mu = mu_\mu + eA_\mu$, agrees with Lect. 13.
Word of caution: When we construct the variation of trajectory $x^\nu \rightarrow x^\nu + \delta x^\nu$, should consider that all δx^ν are independent, and should consider $\dot{x}^\nu \dot{x}_\nu \equiv u^\nu u_\nu \neq 1$ when constructing variations.

Problem 2 (solution)

- ① The lagrangians $\tilde{\Lambda}, \bar{\Lambda}$ differs from Λ only in the first term, so we will repeat the discussion of Lecture 13 only for this contribution:

$$\tilde{\Lambda}_0 = \frac{mc^2 \dot{x}_\mu \dot{x}^\mu}{2}, \quad \delta \tilde{\Lambda}_0 = \frac{mc^2}{2} 2\eta_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{d\delta x^\nu}{d\tau} \equiv mc^2 u_\nu \frac{d\delta x^\nu}{d\tau}$$

and this result agrees with what we got in Lecture 13 for variation of the kinetic term. We may construct Euler-Lagrange equations:

$$\frac{d}{d\tau} \mathcal{P}_\mu = m \frac{du_\mu}{d\tau} + e \frac{dA_\mu}{d\tau} = m \frac{du_\mu}{d\tau} + e \frac{\partial A_\mu}{\partial x^\nu} \dot{x}^\nu = m \frac{du_\mu}{d\tau} + e \frac{\partial A_\mu}{\partial x^\nu} u^\nu$$

$$\frac{\partial \tilde{\Lambda}}{\partial x^\mu} = eu^\nu \frac{\partial A_\nu}{\partial x^\mu}$$

so Euler-Lagrange equation yields (in agreement with Lecture 13):

$$\frac{d}{d\tau} \mathcal{P}_\mu - \frac{\partial \tilde{\Lambda}}{\partial x^\mu} = 0 \Rightarrow m \frac{du_\mu}{d\tau} + e \underbrace{\left(\frac{\partial A_\mu}{\partial x^\nu} - \frac{\partial A_\nu}{\partial x^\mu} \right)}_{F_{\nu\mu} = -F_{\mu\nu}} u^\nu = 0$$

Problem 2 (solution)

- ① Similarly, for the “kinetic” (field-independent) term $\bar{\Lambda}$ can see that for any function f , the variation is

$$\delta f(\dot{x}_\mu \dot{x}^\mu) = \left. \frac{\partial f(y)}{\partial y} \right|_{y=\dot{x}_\mu \dot{x}^\mu=1} \dot{x}_\nu \delta \dot{x}^\nu$$

so if $\left. \frac{\partial f(y)}{\partial y} \right|_{y=1} = \frac{1}{2}$, then the lagrangian $\bar{\Lambda}_0 = mc^2 f(\dot{x}_\mu \dot{x}^\mu)$ will give the same result as $\tilde{\Lambda}_0$, so we'll get the same equation

- ② For analysis of nonrelativistic limit, we should take into account that $\dot{x}_\mu \dot{x}^\mu = u_\mu u^\mu = 1$, and

$$L_{\text{nonrel}} = (\text{const}) + \Lambda d\tau/dt = (\text{const}) + \Lambda \sqrt{1 - v^2/c^2}$$

so after substitution $\Lambda \rightarrow \tilde{\Lambda}$ and expansion over v , we may get

$$L_{\text{nonrel}} = (\text{const}) + \frac{mv^2}{2} + e\varphi - e\mathbf{v} \cdot \mathbf{A}$$